

ARM-450 rev I.1 -- log summary (release pass 2026-09-27, J3_p1 lug fix)

ALL 25 CHECKS AT 0 FAILURES -- print verdict GREEN (fit test first; the 6 servo housings after coupon 07 fits your servo).

#	check	what it proves	result	key numbers
1	Drive trains	J1 .. J6: servo seat, horn on the axis, coupler, bearings, shaft, every part of the joint against every other, servo insertion path	0 failures	23 + 21 + 23 + 16 + 46 individual checks passed (J1, J2, J3, J4, J5/J6)
2	Bearings and bolts as solids	every bearing seated on its pockets, every bolt through real holes	0 failures	54 checks passed
3	Link lock	J2 / J3 links on their shafts: M5 set screws on the double-D flats, centring rings	0 failures	2 x M5 x 10 per link half, all tightened on a flat
4	Every fastener	each screw passes free, bites >= its minimum, head seated, tip free; inserts in pockets	0 failures	114 screws, 61 inserts, 4 nuts against 49 parts
5	Parameters in the files	every key dimension measured back out of the exported print files	0 failures	265 measured, 0 off
6	Tool access	a hex key / screwdriver reaches every screw at the build step where it is fitted	0 failures	114 screws: 78 always clear, 36 clear at their step
7	Rule 4: mate, never overlap	all pairs of the whole assembly, exact Boolean volume; only designed pathways (bearing race, horn, screw); every body attached; 3 injected faults must be caught	0 failures	232 components; 3 of 3 faults detected
8	Gravity springs	spring points, lengths and forces; J2 / J3 torque with the springs	0 failures	worst case arm level: J2 SF 1.07, J3 SF 1.26 (ST3215 0.88 N.m continuous); no payload
9	Whole-arm sweep	each joint through its range, every part against every other; injected block caught	0 failures	joints with a clash: 0; control DETECTED
10	6 degrees of freedom	each axis measured twice (servo horn = bearing pockets), Jacobian rank	0 failures	rank 6 in the working range; spherical wrist (axes meet within 0.000 mm)
11	Manufacturer's servo	Waveshare's own ST3215 model (rear idler horn) in all 6 seats; servo screws in its holes; driven chains shifted 0.20 mm (its horn)	0 failures	servo screw cores in the real holes 20 / 20; worst joint clearance 0.502 mm
12	Take it apart	each servo unit slid out along its path, exact Booleans, our servo AND the manufacturer's	0 failures	7 of 7 paths free, both servo models
13	Joints moving together	450 combined poses of J1 .. J5: every pair of bodies	0 failures	every pair: neighbours >= 0.47 mm (running gaps), all others >= 15.0 mm
14	Tipping	centre of mass over the poses vs the foot	0 failures	COM up to 118.0 mm out vs 64 mm foot: foot MUST be fixed (4 table holes)
15	Thin walls / bed	inward rays on every printed part (< 0.8 mm); bed size	0 failures	thin walls: 0 part(s); all parts fit a 180 mm cube
16	Print orientation gates	each part sliced in its chosen print orientation: islands, sub-layer features	0 failures	0 islands; the 2 'MARGINAL z 16.1' notes are the seam lip's top face (2.0 mm wide below)
17	Static all-pairs	second, independent method: sampled points + penetration depth	0 failures	39 parts, 17 touching pairs, 0 clashes
18	Wiring	both bus plugs into every servo (sockets from the maker's model + your photo), way out for wires	0 failures	nearest wall 1.98 mm; every lead has a way out
19	Screw heads vs turning parts	every screw beside a joint, joint turned -180 .. +180 deg: clearance in range, first contact angle	0 failures	J3 fork heads 4.32 mm (contact at 94 deg), J5 6.85 mm, caps 1.30 mm
20	Union joints	every glued feature of every printed part joined over >= 0.80 of its own section (rev I.1, after the J3 lugs were found on a 0.1 mm sliver)	0 failures	0 weak joint(s); the J3 lugs now 136 mm2 (were 34)
21	Plug slots	each bus plug pulled out of its socket with the arm ASSEMBLED: straight through a slot, or straight then sideways out	0 failures	0 of 12 plugs cannot; routes in PLUG_INSERT.log
22	Cable route	one joint per cable; path change over the range; loop to leave	0 failures	largest path change 42 mm; loops 31 .. 61 mm
23	Sections (manual check)	every joint sliced through its axis, each interface measured	0 failures	0 clashes in the sections
24	Print slicer	the print-ready STLs themselves sliced	0 failures	0 islands, 0 unprintable layers
25	Assembly	the full arm exported as STEP	0 failures	232 components (parts, servos, bearings, every screw, insert, spring)

Important PDFs -- which one to open for what

ARM450_AUDIT_REPORT.pdf 38 pages
READ FIRST: the verdict, what was found and fixed, your 4 snapshot questions answered with pictures, what is still to do

ARM450_REV_I_FINAL.pdf 48 pages
every joint as exact sections (overlap in red, contact green, gaps measured), the whole arm, and every check log in full

ARM450_POSES.pdf 6 pages
the full assembly in 6 poses, clearance measured in each

ARM450_PARTS.pdf 54 pages
every printed part alone: 4 views + sections of its features

ASSEMBLY_MANUAL/ARM450_ASSEMBLY_MANUAL.pdf 32 pages
the build, one page per step (added parts in red), every screw's size and place, cable loops, service notes

ARM450_REV_I.pdf 8 pages
the six joints, one page each: fixed side, driven side, servo, how it is held

RULE5_VIEWS/ARM450_INTERFACES.pdf 19 pages
the Rule 5 interface sections on their own (also inside the FINAL pdf)

LOG_SUMMARY.pdf 2 pages
this summary

REV_H_VERIFICATION.pdf 1 pages
SUPERSEDED rev H report: now a pointer page (original in archive_superseded/)

Where they are: REV_H/ on this machine, RELEASES/2026-09-27_rev_I1/PDF/, and PDF/ in ARM450_rev_I_print on your pen drive. Full logs: REVI_CHECKS.log and FINAL_RUN.log in REV_H/.